December 2025 CSCA Chemistry

Atomic number H: 1 O: 16 Na: 23 Cu: 64

Select this question (there are 4 options per question, and only 1 is correct)

1. Which of the following changes is a chemical change? (...)

   A. Iron rusts B. Activated carbon adsorption of pigments

   C. volatilization of concentrated hydrochloric acid D. Ice melts into water.

2. The following compounds are clearly correct (...)

   A.C0 Carbon dioxide B.$FeCO_3$ ferrous carbonate

   C.Na_2CO_3 Baking soda D.CuCl – Copper chloride

3. Which of the following compound names is incorrect?

   A.$CH_3CH_2$0H Ethanol B.CH_2=CH_2 ethylene

   C.HC Three HC Acetylene D.CH_3CH_3 ethane

4. Which of the following chemical terms is correct?

   A.2 One hydrogen molecule: 2 H B.4 Water molecules:H_2O_2

   C.5 sodium ion: $5Na^+$ D.3 Oxygen atoms 3O

5. Which of the following substances are pure substances?

   A. hydrochloric acid B.Ice-water mixture

   C. air D.Saturated saltwater

6. A group of substances that are allotropes of each other is ()

   A. White phosphorus and red phosphorus B. Ice and dry ice

   C. Water and hydrogen peroxide D. Oxygen and liquid oxygen

7. The oxidation number of N in NH_4NO_3 is ()

   A. -3, +3 B. -3, +5

   C.+1, +5 D.+1, -5

8. Which of the following substances is a solution? (...)

   A. distilled water B. normal saline

   C. arachis oil D. cement paste

9. When the flame color reaction is performed with sodium chloride, the flame appears (...).

   A. pale asphyxia B. yellow

   C. nattier blue D. green

10. Among the following substances, which one is a non-electrolyte?

A. natrium aceticum B. saccharose

C. caustic soda D. sulphuric acid

11. Which of the following substances is an electrolyte?

A.NaCl B. aluminium metallicum

C. saccharose D. carbon dioxide

12. Which of the following reactions is an oxidation-reduction reaction?

A.$CaCO_3 \Delta CaO + CO_2\uparrow$ B.$HCl + NaOH \rightarrow NaCl + H_2O$

C.$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$ D.$CO_2 + H_2O \rightarrow H_2CO_3$

13. Among the following substances, the one commonly used as an oxidizing agent is ().

A. iron B. carbon monoxide

C. mineral chameleon D. hydrogenation

14. Which of the following groups are allotropes of each other?

A. All-inclusive stone and stone within stone

B.CH_4 and C_4H_8

D.CO and C0 2

C.^{12}C and ^{14}C

15. Which of the following compounds has an acidic aqueous solution?

A.NaCl B.NaOH

C.Na_2CO_3 D.NH_4Cl

16. When phenolphthalein solution is added to the solution, if it turns red, the solution is ().

A. basicity B. neutral

C. acidity D. unidentified

17. Which of the Following Gases Can Be Collected Using Both the Drainage Method and the Upward Air Displacement Method?

A. hydrogen B. oxygen

C. carbon dioxide D. methane

18. From a 1 L 1 Mol·L^{-1} NaOH Solution, 50 ML of NaOH Solution Is Removed. The Concentration of the 50 ML NaOH Solution Is ().

A.0.02$mol \cdot L^{-1}$ B.1 $mol \cdot L^{-1}$

C.0.2 $mol \cdot L^{-1}$ D.2 $mol \cdot L^{-1}$

19. Which of the following statements about the use of substances is incorrect?

A.02 assisted combustion B.CO can be used to put out fires.

C. Carbon can be used as fuel D.75% alcohol can be used to disinfect 20.^{16}S position in the periodic table of elements ()

A. Week 3, Group IVA B.The 3rd cycle, VIA group

C. Week 2, Group IVA D.The 2nd cycle, VA group

21. When a cup of concentrated sulfuric acid is left in the air, what changes occur after a period of time?

A. mass reduction of solvent

B.The mass of the solution has increased.

C. increase in the mass of solute

D.The mass of the solution has decreased.

22. Which of the following groups of compounds are homologues?

A.CH_3CH_3 and HC three CH B.CH_3 CH=CH_2 and CH_3CH_3

C.HC_3 CH=CH_2 and CH_3 C =CCH_3 D.CH_3CH_3 and $CH_3CH_2CH_3$

23. Which of the following reaction types is correctly described?

A.MnO_2+ 4HCl (concentrated) → $MnCl_2 + Cl_2 \uparrow + 2H_2O$ Chemical reaction

B.Fe_2O_3+ 3CO → 2Fe +$3CO_2$ Substitution reaction

C.$3Fe + 2O_2 \rightarrow Fe_2O_4$ decomposition reaction

D.HClO + NaOH → NaClO +H_2 O double decomposition reaction

24. When the reaction reaches equilibrium, the constant in the reaction system is ().

A. A. The atom exists in the product molecule of the reactant molecule.
B. The time of reactant concentration and product concentration.

C. The rate of the reverse reaction is the reciprocal of the forward reaction rate. D. Mass of reactants and mass of products

25. Which of the following substances is an oxide?

A.H_2O B.NaOH

C.O_2 D.$KClO_3$

26 Which of the following statements about catalysts is incorrect?

A. A. The chemical properties of the catalyst remain unchanged before and after the reaction. B. The mass of the catalyst changes before and after the reaction.

C. Certain reactions can utilize different catalysts. D. Catalysts can alter the reaction rate

27. Which of the following groups contains the same number of chemical bonds?

A.NaCl and HCl B.O_2 and H_2

C.HNO_3 and NH_4NO_3 D.NaOH and KCl

28. Which of the following glassware is called a beaker?

A 

B. 

C. 

D 

29. When $BaCl_2$ solution is added to a solution, a white precipitate is formed, and the precipitate does not dissolve when nitric acid is added. The following statement is correct is ()

A. The solution must contain S024^{-}

B.The solution may contain CO23^{-}

C. The solution must contain Ag^{+}

D. The solution may contain S024^{-}or Ag^{+}30. The reaction of hydrochloric acid and sodium hydroxide belongs to ()

A. decomposition reaction B. neutrallzation reaction

C. combination reaction D. displacement reaction

31. Which of the following groups of elements belongs to the same group?

A. Hydrogen, Lithium, Sodium B.Carbon, oxygen, sulfur

C. Hydrogen, carbon, nitrogen

D.Sodium, magnesium, aluminum

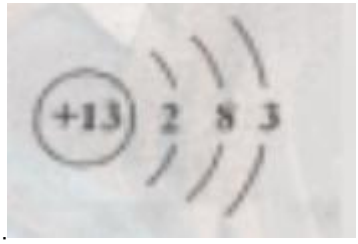
32. Which of the following is the correct atomic radius gradient rule for allicin?

A.B<Al<Mg<Na B.Al<B<Mg<Na

C.Al<B<Na<Mg D.B<Al<Na<Mg

33. The Electron Configuration Diagram of the Na (sodium) Element Is ()

A. 

B. 

C 

D 

At 34.25°C, the pH of 100mL of 0.05 mol·L $^{-1}$ H_2SO_4 solution is () after the water is evaporated to concentrate it to 10 mL.

A. 1 B. 2

C. 14 D. 0

35. In the reaction $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ in a 2L closed container, the amount of reactant SO_2 decreases from 1 mol to 0.6 mol within 10s. The reaction rate v(O2) is ().

A.0.02mol· $L^{-1} \cdot s^{-1}$ B.0.01mol· $L^{-1} \cdot s^{-1}$

C.0.2mol· $L^{-1} \cdot s^{-1}$ D.0.04mol· $L^{-1} \cdot s^{-1}$

36. The reagent used to detect Cl‧ ions is ().

A.$AgNO_3$ and dilute sulfuric acid B·$BA(NO)_2$ and dilute nitric acid

C.$BaCl_2$ and dilute hydrochloric acid

D.$AgNO_3$ and dilute sulfuric acid

37 After the following reactions reach equilibrium, reducing pressure or increasing temperature can shift the chemical equilibrium toward the reaction direction. A.$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ is an endothermic reaction. B.$2NO_2(g) \rightarrow N_2 + O_4$ is an exothermic reaction. Should

C.$3O_2(g) \rightarrow 2O_3$ The forward reaction is an endothermic reaction D.C (s) +$CO_2(g) \rightarrow 2$ CO (g) Both directions

should be an endothermic reaction

38. Which of the following chemical reactions is a substitution reaction?

A. Reaction of ethylene with hydrogen chloride

B. Reactions of Bismuth with Hydrogen

C. Reaction of methane with chlorine

D.Ethylene reacts with water

39. Which of the following statements about organic compounds is correct?

A. Organic compounds contain only carbon and oxygen elements. B. Organic compounds cannot be synthesized through artificial methods.

C. Organic compounds exhibit isomerism. D. Compounds containing carbon elements can be inorganic substances or

Therefore, it is organic matter.

40. When using solar energy to decompose water for hydrogen production, if 0.02mol of water is decomposed, the following statement is correct.

A. The volume of H_2 generated is 0.224L (standard state). B. The amount of substance of H_2 generated is 0.01mol

C. The amount of substance of H_2 generated is 0.02 mol D.The mass of the generated H_2 is 0.02g

41 The molar concentration of the solution of the predatory solenium is 0.5 mol. L^{-1} is ()

A.1L contains a solution of 40g NaOH B.11.2 L HCl is dissolved in water to prepare a 1L solution

D.20g of NaOH dissolved in 1 L of water

C.1L solution of K2SO4 containing 1 mol of K+

42. For the reaction C (S) +H_2 0 (g) → C0 (g) +H_2(g), the reaction equilibrium shifts in the reverse direction for ()

A. cooling and pressure reduction

B.Heating and pressurizing

C. cooling and blood pressure reduction

D.Cooling and pressurization

43. The liquid that forms a red Cu20 precipitate with newly prepared Cu(OH)2 is ()

A.CH_3COOH B.$CH_3COOCH_2CH_3$

C.CH_3CHO D.CH_3CH_2OH

44. Which of the following ion groups can coexist in large quantities in the specified solution?

A. In alkaline solution: Al^{3+},HCO_3^-, Cl-, K^+B. In acidic solution: Ag^+,NO_3^-, Cl-Na^+

C. In alkaline solution: Ba^{2+}, $CO3^-$, Cl-, K^+D. In acidic solution: Fe^{3+}, $SO2^-$,NO_3^-, Mg^{2+}

45. A mixture of CuO and Cu powder (23.4g) was reduced with CO. After complete reaction, 20.2g of pure copper powder was obtained. Thus, the mass of CuO in the original mixture is ().

A.20g B.32g

C.40g D.16g

46. In a chemistry lab session, the students performed the following procedures to achieve the objective: ()

Operation in experiment	Experimental objective
① Add excess NaOH solution to the $ZnCl_2$ solution	Preparation of zn(0H)$_2$ precipitate
② Heat and evaporate the $AlCl_3$ solution	Wastewater $AlCl_3$ prepared from $AlCl_3$ solution
③ Add dilute nitric acid solution, filter, wash, and dry	Remove the mixed CuO from the Cu content.
④ CO_2 is passed into a C_6H_5 ONa solution and the solution becomes turbid	Comparison of the acidity of carbonic acid and phenol

A.④ Experimental Procedures and Objectives B.① Experimental Procedures and Objectives

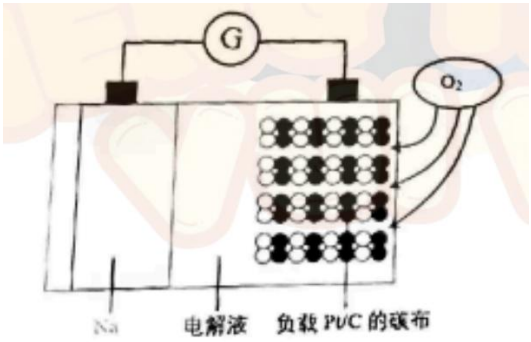
C.③ Experimental Procedures and Objectives D.② Experimental Procedures and Objectives

47 A white powder consists of two substances. To determine its composition, conduct the following experiments: ① Add a small sample to an adequate amount of water; the solid portion dissolves. Then add an adequate amount of dilute hydrochloric acid; bubbles form, and the solid dissolves completely. ② Add a small sample to an adequate amount of dilute sulfuric acid; bubbles form, and after shaking, solid remains. The white powder could be ().

A.$AgNO_3$,Na_2CO_3 B.NaOH,$CUSO_4$

C.$CaCO_3$,$BaCO_3$ D.Na_2CO_3,$BaCO_3$

48. Scientists have developed a room-temperature Na-O_2 secondary battery, using $NaClO_4$ dissolved in an organic solvent as the electrolyte. The carbon cloth of Pt/C metal is used as the electrode material, and the overall reaction of the battery is:O_2+ 2Na → Na_2O_2. The following statement is correct



A. During discharge,ClO_4^-moves toward the positive electrode B. It releases O_2 during discharge and absorbs O_2 during charging. C. During discharge, the negative electrode reaction is: Na^++ e- → Na D. During charging, the positive electrode reaction is:

$$Na_2O_2 - 2e^- \rightarrow 2Na^+ + O_2 \uparrow$$